

REAL-DATA DRIVEN DENOISING AUTOENCODERS FOR 1D PERIODIC TIME-SERIES SIGNALS

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ABSTRACT

This paper studies denoising autoencoders for 1D periodic signals. Real signals rarely come with clean targets. We address that by fitting a simple sinusoid to weekly Wikipedia pageview series and using that fit as a proxy clean signal. We fit three candidate noise models to the residuals, generate a synthetic training set from the fitted signal family, and compare a multilayer perceptron (MLP) autoencoder with a one-dimensional convolutional neural network (1D CNN) autoencoder. The CNN gives the best synthetic result at 15.18 dB signal-to-noise ratio (SNR) improvement with 53,313 parameters, versus 14.49 dB for the MLP with 101,136 parameters. The same CNN gives 8.91 to 11.93 dB improvement on a controlled per-article sine recovery test at the fitted Gaussian noise level ($\sigma = 0.380$ in log-pageview space). On the raw Wikipedia series it gives 2.27 to 8.06 dB improvement against the fitted-sine proxy. These results show that the synthetic benchmark is strong and that the real-data result is useful but limited by the proxy target.

1 INTRODUCTION

A denoising autoencoder learns to recover a clean signal from a corrupted input (Vincent et al., 2008). This is a simple idea when the clean target is known. It is harder on real time-series data, where a true clean target is rarely available.

We use weekly Wikipedia pageviews as a real 1D signal source. Disease-related pages on Wikipedia show strong seasonal structure (Generous et al., 2014). That structure makes them a good fit for this project. We model each series with a single sinusoid plus offset and trend. We then treat the fitted sinusoid as a proxy clean target.

Supervised training itself uses synthetic windows sampled from the fitted signal and noise statistics, so the benchmark still has exact clean targets.

This setup lets us study three related questions. First, can a proxy clean model estimated from real data support a useful synthetic denoising benchmark. Second, under that benchmark, how do a multilayer perceptron autoencoder and a one-dimensional convolutional autoencoder compare in accuracy and efficiency. Third, how robust is the selected model to changes in latent size, noise family, and noise intensity. We also ask how well it transfers back to the original Wikipedia series.

The central tradeoff is clear from the start. Synthetic data gives a true target and clean metrics. Real data is more meaningful, but the target is only a proxy. The report keeps both views.

2 DATA

Source. We use daily pageviews from the Wikimedia REST API (Wikimedia Foundation, 2026). The five articles are *Influenza*, *Common_cold*, *Fever*, *Cough*, and *Christmas*. We freeze the date range at 2018-01-01 to 2026-04-20 for reproducibility. We resample the daily counts to weekly totals. Each series has 472 weekly points.

Why these articles. The four health-related pages show a clear yearly pattern tied to flu season. Christmas is a positive control. Its yearly peak is large and regular. If the pipeline cannot recover that pattern, the method is not working.

Transform. All downstream processing uses $y = \log(1 + \text{pageviews})$. This reduces the scale gap across articles. It also makes the dominant yearly pattern closer to a sine wave.

3 METHOD

3.1 PROXY CLEAN SIGNAL

For each article, we fit

$$\hat{y}[t] = c + mt + A \sin(2\pi f t + \phi). \quad (1)$$

We initialize the frequency with a Hanning-windowed fast Fourier transform (FFT). We then refine all parameters with bounded least squares. The fitted curve is our proxy clean signal. The residual is $r[t] = y[t] - \hat{y}[t]$.

Table 1 shows the fitted periods and coefficient-of-determination values (R^2). Four articles land near the expected one-year period. Cough is the exception. Its fitted period is about 301 weeks, so its proxy target is weaker than the others.

Table 1: Per-article sine-fit summary in log space.

Article	Period (weeks)	R^2	Amplitude
Influenza	51.3	0.633	0.380
Common_cold	51.1	0.589	0.307
Fever	50.9	0.776	0.136
Cough	300.9	0.570	0.186
Christmas	52.1	0.554	0.897

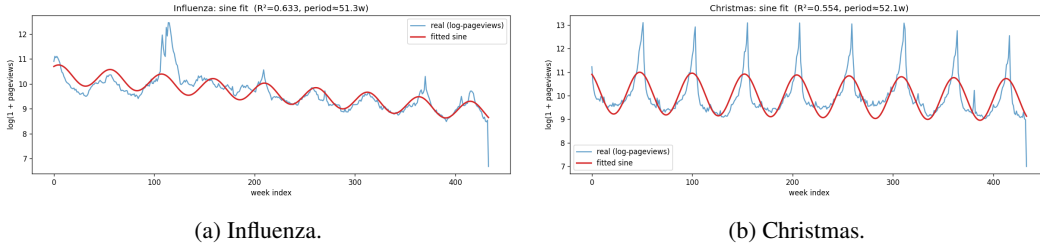


Figure 1: Example sine fits in log space. The global fit captures the annual cycle, but it misses some year-to-year amplitude change.

3.2 RESIDUALS AND NOISE MODELS

We pool residuals across articles and fit three candidate noise models.

- **Gaussian noise:** additive zero-mean noise with standard deviation σ .
- **Masking noise:** short contiguous segments are replaced by the local mean.
- **Impulse noise:** sparse spikes are added at random indices.

We score each fitted noise model against the observed residuals. The main score is Wasserstein distance. We also report the Kolmogorov-Smirnov statistic, the kurtosis gap, and the standard-deviation ratio. Lower Wasserstein is better.

3.3 SYNTHETIC DATASET

We turn the fitted real-data parameters into a synthetic signal family. We sample offsets, slopes, amplitudes, frequencies, and phases from the fitted ranges. Each clean window has length 128. We z-score each clean window. We then corrupt it with one fitted noise model.

Each training split uses 8000 train windows, 1000 validation windows, and 1000 test windows. This gives a true clean target for every synthetic sample. That makes mean squared error (MSE) and signal-to-noise ratio (SNR) easy to interpret.

3.4 MODELS, TRAINING, AND METRICS

We compare two models.

- **MLP autoencoder:** $128 \rightarrow 256 \rightarrow 64 \rightarrow z \rightarrow 64 \rightarrow 256 \rightarrow 128$.
- **1D CNN autoencoder:** encoder channels $1 \rightarrow 16 \rightarrow 32 \rightarrow 64$ with kernel sizes 7, 5, and 3 and stride 2. The decoder mirrors this stack with transposed convolutions.

We train with mean squared error, Adam, learning rate 10^{-3} , batch size 64, and early stopping. The default latent size is $z = 16$ unless noted otherwise.

We report

$$\text{MSE}(x, \hat{x}) = \frac{1}{NL} \sum_{n,t} (x_n[t] - \hat{x}_n[t])^2, \quad (2)$$

$$\text{SNR}_{\text{dB}}(x, \hat{x}) = 10 \log_{10} \left(\frac{\|x\|_2^2}{\|x - \hat{x}\|_2^2} \right), \quad (3)$$

$$\Delta \text{SNR} = \text{SNR}_{\text{dB}}(x, \hat{x}) - \text{SNR}_{\text{dB}}(x, x_{\text{noisy}}). \quad (4)$$

4 EXPERIMENTS AND RESULTS

4.1 EXP 2: NOISE MODEL SELECTION

Table 2 shows the fitted noise scores. Gaussian noise is the best match by Wasserstein distance. It also matches the residual standard deviation almost exactly; the fitted Gaussian standard deviation on the pooled residuals is $\sigma = 0.380$ in log-pageview space. The main weakness is tail fit. The real residuals are more heavy-tailed than a Gaussian.

Table 2: Noise-model selection on pooled residuals. Lower Wasserstein is better.

Kind	Wasserstein	Kolmogorov-Smirnov	Kurtosis diff	Std ratio
Gaussian	0.106	0.118	-10.06	0.999
Masking	0.241	0.569	819.49	0.111
Impulse	0.275	0.559	48.97	1.908

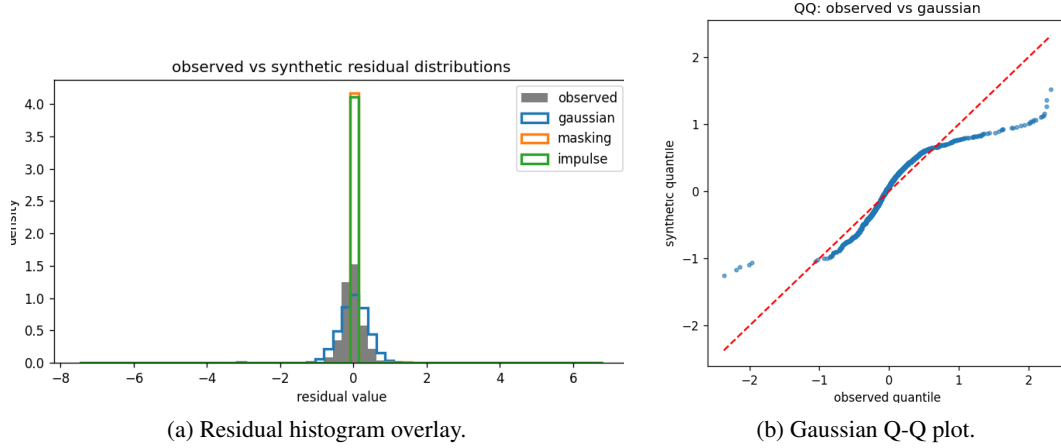


Figure 2: Residual fit diagnostics. Left: Gaussian matches the center of the distribution well; masking and impulse are visibly more distorted. Right: the center tracks the diagonal, but the tails bend away, which shows the residuals are more heavy-tailed than a Gaussian.

4.2 EXP 3: ARCHITECTURE COMPARISON

Table 3 compares the MLP and the 1D CNN on the synthetic benchmark. The 1D CNN is better on every main test metric. It improves SNR by 15.18 dB. The MLP improves SNR by 14.49 dB. The CNN also uses about half as many parameters. The price is training time. The CNN is much slower to train.

Table 3: Architecture comparison on the synthetic test set.

Model	Params	Test MSE	Output SNR	Δ SNR	Train (s)
MLP	101,136	0.00632	22.95	14.49	56.8
CNN1D	53,313	0.00525	23.63	15.18	585.7

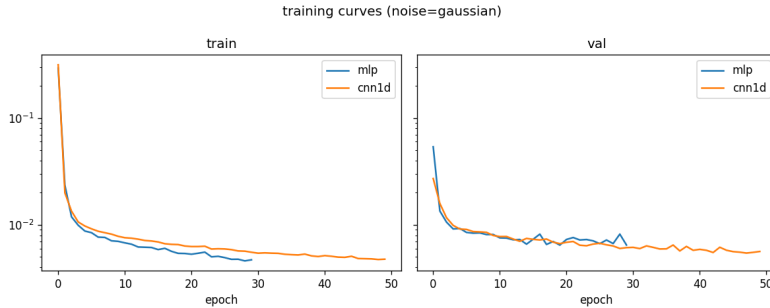


Figure 3: Training and validation loss for the architecture comparison. Both models converge cleanly. The CNN reaches the lower final validation loss.

4.3 EXP 4: LATENT DIMENSION SWEEP

Table 4 shows the latent sweep for the CNN. The big jump is from latent 4 to latent 8. The best test MSE is at latent 16. Latent 32 gives almost the same SNR improvement, but it is larger. We use latent 16 for the rest of the report.

Table 4: Latent sweep for the 1D CNN.

Latent	Params	Test MSE	Δ SNR
4	28,725	0.01407	11.56
8	36,921	0.00559	14.91
16	53,313	0.00527	15.17
32	86,097	0.00535	15.19
64	151,665	0.00534	15.16

4.4 EXP 5 AND EXP 6: ROBUSTNESS TO NOISE TYPE AND NOISE LEVEL

Cross-noise robustness shows a clear diagonal pattern. Each model works best on the same noise type it saw in training. The masking-trained and impulse-trained models do very well on matched noise, at 25.17 dB and 24.55 dB respectively. Their transfer is weaker. The worst case is masking-trained on impulse noise at -2.14 dB. The Gaussian-trained model is the most balanced. It gets 15.13 dB on Gaussian, 11.07 dB on masking, and 9.74 dB on impulse.

Noise-level generalization is also strong. The CNN is trained at the fitted Gaussian level ($\sigma = 0.380$). From $0.75\times$ to $3.0\times$ that level, SNR improvement stays between 14.34 and 15.73 dB. Performance drops at $0.25\times$ because the input is already much cleaner.

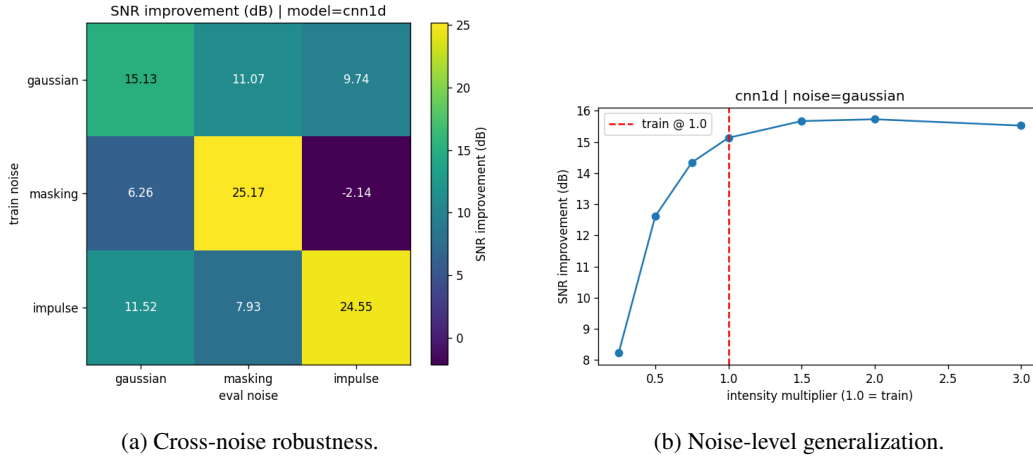


Figure 4: Generalization results for noise type and noise level. The left panel shows strong diagonal dominance. The right panel shows stable performance from $0.75\times$ to $3.0\times$ noise.

4.5 EXP 7: RAW WIKIPEDIA EVALUATION

This experiment works directly on the raw Wikipedia series. We slide 128-week windows with stride 1. We z-score each window, denoise it, undo the z-score, and overlap-average the window outputs back into one full series.

We then compare the reconstruction to the fitted sine proxy. Table 5 shows the results. Improvement ranges from 2.27 to 8.06 dB. Christmas is the strongest case. Cough is the weakest. That is consistent with the earlier fit quality.

Table 5: Raw Wikipedia evaluation against the fitted-sine proxy.

Article	MSE vs fitted sine	Reconstruction SNR	Raw-series SNR	Δ SNR
Influenza	0.0855	30.37	27.74	2.63
Common_cold	0.0656	31.06	28.41	2.65
Fever	0.0160	37.12	32.96	4.16
Cough	0.0493	30.99	28.72	2.27
Christmas	0.0502	32.99	24.94	8.06

4.6 EXP 8: CONTROLLED PER-ARTICLE SINE RECOVERY

This experiment is the closest real-data match to the synthetic task. For each article, we take its fitted sine as the clean target. We add Gaussian noise at the fitted level ($\sigma = 0.380$). We then denoise it with the trained CNN. The mean improvement is 10.07 dB. The range is 8.91 to 11.93 dB.

The clean-pass SNR is also strong. When we feed the clean fitted sine into the autoencoder, the mean SNR is 65.46 dB. That tells us the model adds very little distortion when the input is already clean.

Table 6: Per-article sine recovery at the fitted Gaussian noise level ($\sigma = 0.380$).

Article	Reconstruction MSE	Input SNR	Reconstruction SNR	Δ SNR	Clean-pass SNR
Influenza	0.0138	28.07	38.37	10.30	64.77
Common_cold	0.0150	27.61	37.55	9.94	66.49
Fever	0.0174	27.53	36.80	9.26	68.43
Cough	0.0190	26.30	35.21	8.91	71.19
Christmas	0.0095	28.38	40.31	11.93	56.44
Mean	0.0149	27.58	37.65	10.07	65.46

The intensity sweep is stable here too. Across all five articles, the improvement stays positive across the full $0.25\times$ to $3.0\times$ range. At $3.0\times$ noise, every article still gets at least 8.82 dB improvement.

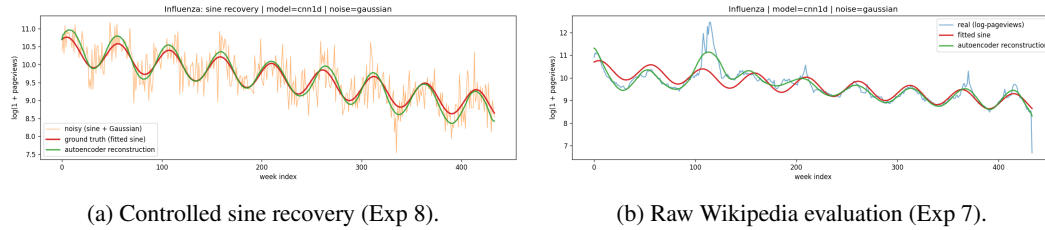


Figure 5: Two views of real-data behavior for Influenza. The left panel matches the synthetic task closely. The right panel shows the harder raw-series setting.

5 DISCUSSION

The synthetic benchmark gives the clearest performance signal. It has a known target. It also matches the way the models are trained. Under that setup, the 1D CNN is the best model we tested.

The raw Wikipedia result is smaller for a clear reason. The fitted sine is only a proxy target. The denoised series can follow real nonstationary structure, such as changing seasonal amplitude, and still look worse against one global sine. That is why Exp 7 and Exp 8 should not be read as the same metric.

Cough is the main edge case. Its fitted period is far from one year. That weakens both the synthetic signal family and the proxy target for raw evaluation. The model still helps, but the interpretation is less clean.

The noise model is another limitation. Gaussian noise wins among the three tested families, but it does not match the residual tails well. A broader noise family could improve realism. The clean signal family is also narrow. It uses one dominant sinusoid plus a linear trend. A richer family could capture more local structure.

6 CONCLUSION

We built an end-to-end denoising autoencoder pipeline for 1D periodic signals. We grounded the synthetic benchmark in statistics from a real dataset. The 1D CNN gave the best synthetic result, the best final model, and stable performance across noise types and levels. It also gave useful gains on two real-data evaluations.

The main caveat is the proxy clean target. On real data, the fitted sine is only an approximation. Even so, the pipeline gives a consistent picture. The synthetic benchmark is strong, the controlled sine recovery test is also strong, and the raw-series result is positive but more conservative.

Code and report files are available at <https://github.com/IslamTayeb/denoise-autoencoder>.

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REFERENCES

- Nicholas Generous, Geoffrey Fairchild, Alina Deshpande, Sara Y. Del Valle, and Reid Priedhorsky. Global disease monitoring and forecasting with Wikipedia. *PLOS Computational Biology*, 10(11): e1003892, 2014.
- Pascal Vincent, Hugo Larochelle, Yoshua Bengio, and Pierre-Antoine Manzagol. Extracting and composing robust features with denoising autoencoders. In *Proceedings of the 25th International Conference on Machine Learning (ICML)*, pp. 1096–1103, 2008.
- Wikimedia Foundation. Wikimedia REST API: Pageviews. https://wikimedia.org/api/rest_v1/, 2026. Accessed: 2026-04-20.